Week 3: Linux Logs, Sysmon, auditd and Snort Setup

Theory Notes, Developing Basic Understanding in SOC

Question 6. What is the difference between signature-based and anomaly-based detection? Give examples of each from a SOC perspective.

Signature-based detection

In this method, the system compares activity with a list of patterns that are already known to be bad. A pattern can be a file hash, a piece of text in a log, or a rule. If something matches, an alert is raised.

- It is quick and very accurate for known threats.
- It is easy to explain, because I can see exactly which pattern matched.
- The weakness is that it cannot catch something new, and a small change made by the attacker can be enough to slip past it.

Anomaly-based detection

Here the system first learns what normal activity looks like, which is called a baseline. Later, anything that moves far away from that baseline is flagged as unusual.

- It can find new or unknown attacks, and unusual behaviour from users inside the company.
- The weakness is that it gives more false alerts, because something new but harmless also looks unusual.
- It also needs time and good data to build a proper baseline.

Side by side

Point	Signature-based	Anomaly-based
Works on	Known patterns	Differences from normal behaviour
Unknown threats	Cannot detect them	Can detect them
False positives	Low	Higher
Effort to set up	Low, because rules are easy to get	High, needs a baseline and tuning
Explaining an alert	Easy	Harder
Updates	New signatures are needed regularly	Baseline needs to be refreshed over time

Examples from a SOC

Type	Example
Signature	A Snort rule such as ICMP Ping Detected, or the Nmap SYN scan rule
Signature	Antivirus detecting a file by its hash
Signature	A SIEM rule that fires when Windows Event ID 4625 repeats many times
Signature	A Sigma or Wazuh rule that spots `mimikatz` in a command line

Type	Example
Anomaly	A user who always logs in from India suddenly logging in from another country at night
Anomaly	A workstation sending far more data than usual to an unknown IP address
Anomaly	A host making many more DNS requests than normal, which may point to DNS tunnelling
Anomaly	UEBA tools that give a risk score to unusual user behaviour

In a real SOC both are used together. Signatures give quick and reliable detection of known attacks, and anomaly detection covers the new and unusual things that signatures miss.